

Short-term synaptic plasticity

Computational Neuroscience

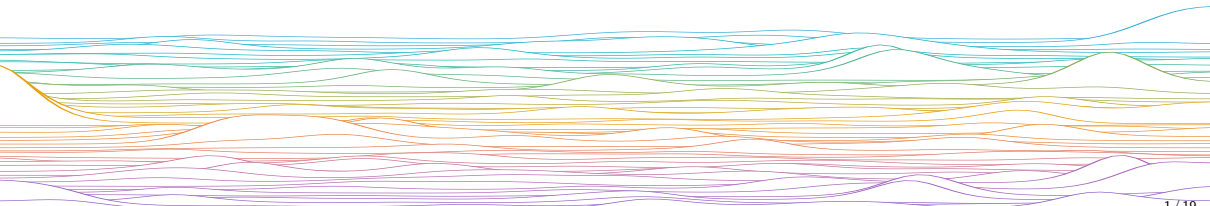
University of Bristol

Adapted from B Mizusaki's slides

M Rule

Learning outcomes:

- ▶ Describe the concept, mechanisms of Short-Term Plasticity (STP)
- ▶ Be able to **write down** the equations for a simple model of STP
- ▶ Be able to describe STP as an information filter.



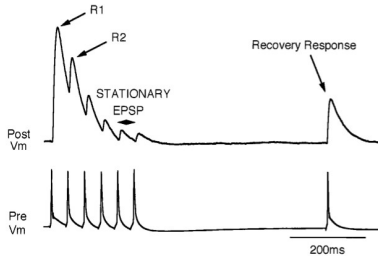
Short-term plasticity

Short-term synaptic plasticity is thought to be involved in fast cognitive processing.

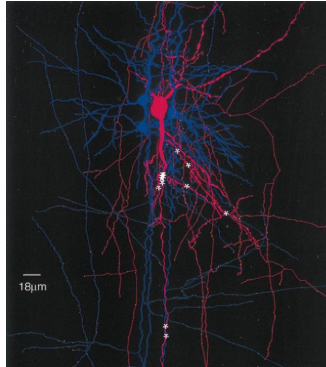
- ▶ Similar timescale to electrical dynamics
 - Short-term synaptic facilitation (strength increases with use)
 - Short-term synaptic depression (strength decreases with use)

STP in an experiment

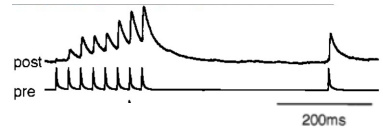
Depression



Tsodyks & Markram, PNAS (1997)



Facilitation



Markram, Wang, & Tsodyks, PNAS (1998)

High- vs. Low-pass filter confusion note

High- vs. Low-pass filter confusion

In lecture, there was some confusion about associating depression, facilitation with high-pass or low-pass filtering in the signal processing sense. Previous iterations of this course claimed that STD is a *low-pass* filter. The lecturer found this counterintuitive, and flipped the associations of high/low pass filtering. This was a mistake (or was it?).

High- vs. Low-pass filter confusion

Fortune & Rose (2001) Opinion: LTD is a low-pass filter appears as settled dogma

Recent models and intracellular data suggest that short-term synaptic depression accounts for a significant portion of this low-pass temporal filtering.

Rosenbaum, Rubin, Doiron (2012): LTD with stochastic release is a high-pass filter

...a synapse model with stochastic vesicle dynamics suppresses information encoded at lower frequencies more than information encoded at higher frequencies, while a model that ignores this stochasticity transfers information encoded at any frequency equally well.

well then...

High- vs. Low-pass filter confusion

Perhaps equating STD/STF to linear low-/high-pass filtering is too simplistic, but we can say this:

STD

- ▶ Attenuates the strength of inputs with many spikes (high firing rates)
 - Low-pass steady-state response
- ▶ Responds more after an abrupt increase in the firing rate
 - High-pass response to positive steps in rate

STF

- ▶ Attenuates the strength of inputs where spikes are rare (low firing rates)
 - High-pass steady-state response
- ▶ Delayed (ramping up) response to an abrupt increase in firing rate
 - (if presynaptic firing rates are high enough for Ca^{2+} concentrations to sum)
 - Low-pass response to positive steps in rate

High- vs. Low-pass filter confusion note

- ▶ Thankfully, the high/low-pass interpretation *is not* on the exam.
- ▶ However, building and discussing ODE models of LTD/LTP *might be* on the exam!
- ▶ Treat STD/STF dynamics as they are, without high-/low-pass interpretation

(end high- vs. Low-pass filter confusion note)

Short-term depression

Too many vesicles are used before more of them are prepared to be released

- ▶ Depletion of the readily releasable pool

Rate of vesicle replenishment limits rate of transmission:

- ▶ ~~Low-pass filter~~ see high- vs. Low-pass filter confusion note

Draw what happens with synaptic vesicles and post-synaptic potentials

Short-term facilitation

AP induces local Ca^{2+} influx in the ***presynaptic*** terminal

- ▶ Elevated Ca^{2+} leads to increased release probability
- ▶ Ca^{2+} trigger for vesicle fusion nonlinear (release still requires presynaptic spike)

Time of Ca^{2+} availability in the terminal limits period of available facilitation:

- ▶ ~~High-pass filter~~ see high- vs. Low-pass filter confusion note

Draw what happens with presynaptic calcium concentrations

Short-term depression model

Think of the synaptic efficacy as being determined by the available amount of a resource

$$\begin{aligned}
 \dot{R} &= \rho_{\text{rec}} I - D \cdot R \cdot \delta(t - t_s) \\
 \dot{E} &= -\rho_{\text{inac}} E + D \cdot R \cdot \delta(t - t_s) \\
 I &= 1 - R - E
 \end{aligned}$$

$R \in \{0, 1\}$ fraction “readily” available

- ▶ Decrements by dR on each spike
- ▶ Recharges from I fraction rate $\rho_{\text{rec}} I$
- ▶ $D \in [0, 1] \sim$ portion resource depleted with each spike

$E \in \{0, 1\}$ fraction “exhausted”:

- ▶ Increments by $D \cdot R$ on each spike
- ▶ $E \rightarrow I$ rate $\rho_{\text{inac}} E$

$I \in \{0, 1\}$ fraction “inactive”

- ▶ Recycled, but not ready to use
- ▶ $I \rightarrow R$ at rate $\rho_{\text{rec}} I$

The Dirac delta notation $q \cdot \delta(t - t_s)$ on the RHS of an ODE is shorthand for a jump condition

- ▶ It means that we should add amount q to the integrated variable when we pass through time t_s
- ▶ You may also see notation t^- and t^+ to denote that two values may be associated with a single time point before and after the jump.
- ▶ It is tempting to assume $\delta(t - t_s)$ can always be viewed as an impulse as in the case of linear ODEs
 - Not always true when the condition generating the impulse comes from the ODE's own state.
 - Writing out the jump conditions explicitly might be more clear.

But, neuroscientists love writing spike (or event) sequences as impulse trains $y(t) = \sum_{t_s \in \mathcal{T}} \delta(t - t_s)$.

Here is a first-order exponential decay synapse (q neurotransmitter released)

$$\begin{aligned}g_s(t) &= \bar{g} \cdot s(t) \\ \tau_s \dot{s} &= -s + q \cdot \delta(t - t_s)\end{aligned}\tag{1}$$

We could model a synapse with STD by using R to scale down the neurotransmitter released

$$\begin{aligned}g_s(t) &= \bar{g} \cdot s(t) \\ \tau_s \dot{s} &= -s + R \cdot q \cdot \delta(t - t_s)\end{aligned}\tag{2}$$

What would we change to get a facilitatory model?

Summary

STP can be modelled as a competition for resources

~~STP works as a frequency filter~~

- ▶ ~~STF: Low pass~~
- ▶ ~~STD: High pass~~ [see high- vs. Low-pass filter confusion note](#)

Most neurons have synapses that are either mostly facilitating or depressing

end

